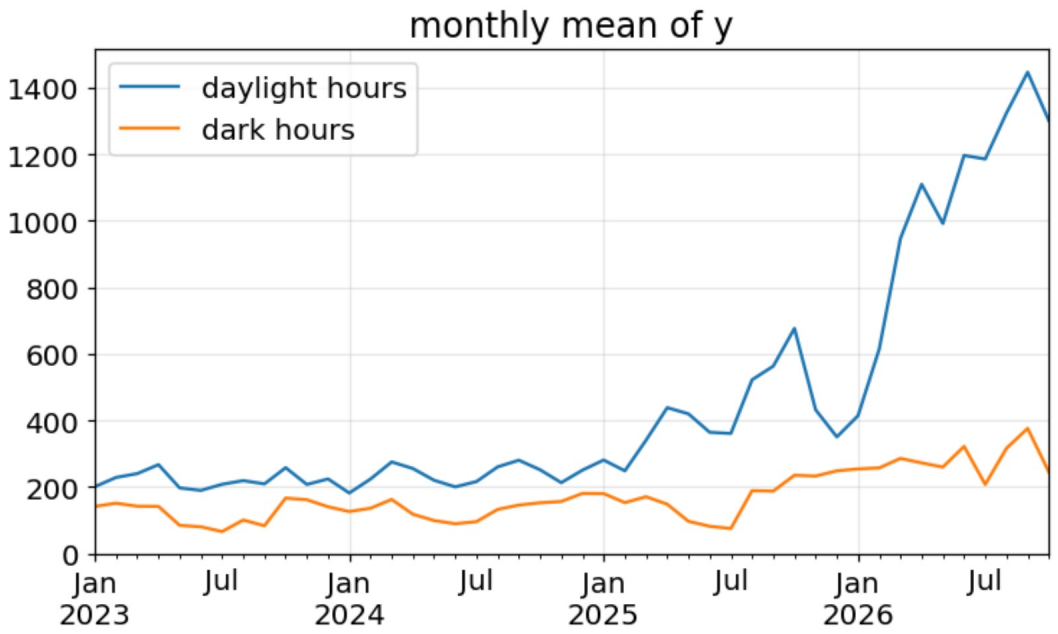
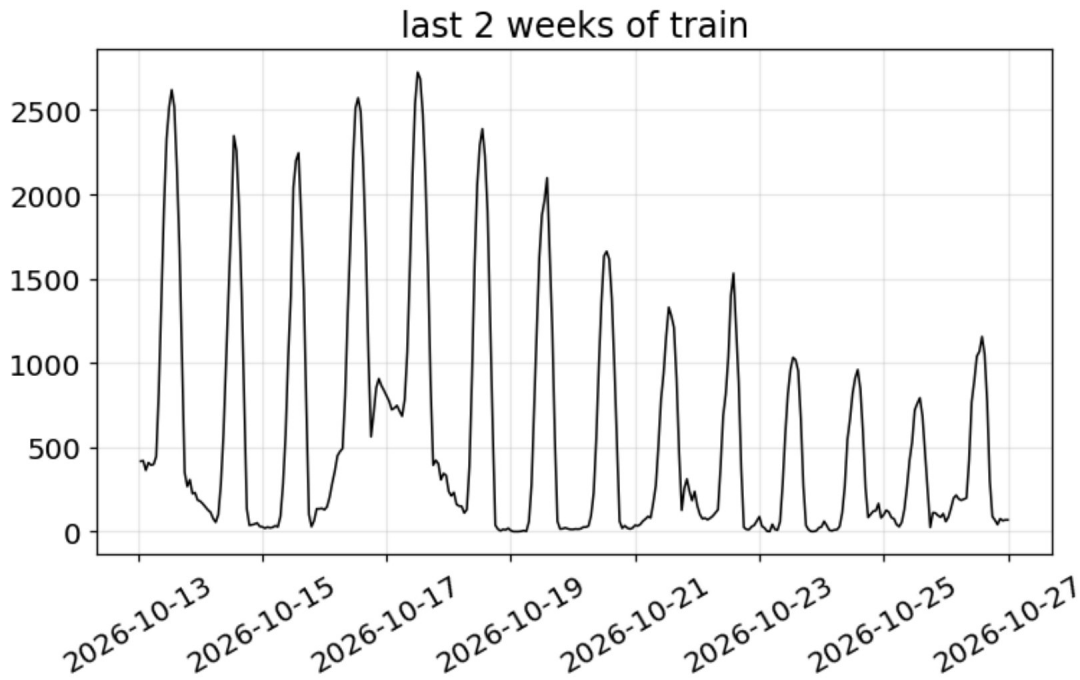
# 48 hours of generation

MATH840 final project: Session 1 vs final forecast

Illia Narbut

# The data




- Solar in the day, wind at night. Output grew about 4x since 2023, mostly solar
- Clock change on 27 Oct: daylight moves from 08-16 to 07-15
- Last week before the forecast: foggy days and almost no wind at night

# Session 1: the model

- LightGBM A: hour, day of year, trend, weather (+ a version with lags 48h / 168h)
- LightGBM B: one model for all horizons, knows the last 6 / 24 / 72 hours
- Forecast = (A + B) / 2
- Interval: 95th percentile of backtest |error| in 10 groups (part of day x day 1/2)

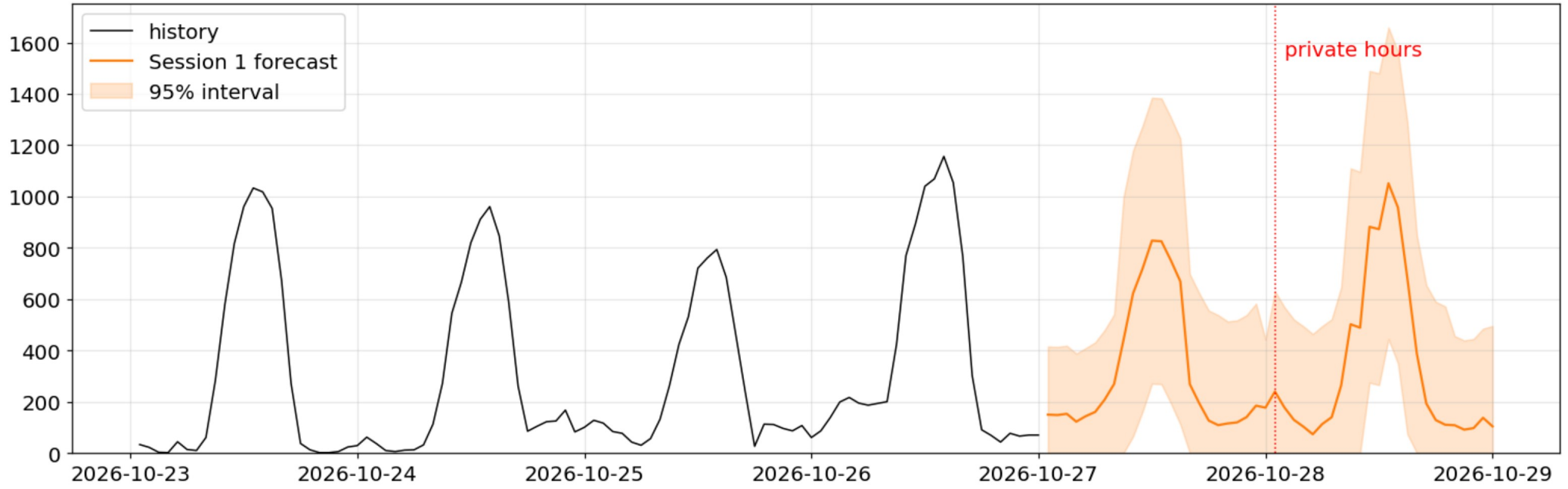
**Backtest, 60 windows**

| | mine | naive |
|---|---|---|
| RMSE | 199.6 | 255.0 |
| MAPE | 87.7% | 133.9% |
| Winkler | 1026 | 1847 |
| coverage | 0.945 | 0.82 |

beat naive in 47 of 60 windows

Problem I saw later: this backtest used the real weather, not a forecast.

# Session 1: public leaderboard (27 Oct)




| | RMSE | MAPE | Winkler | place |
|---|---|---|---|---|
| me | 115.1 | 65.0% | 674 | 19 |
| naive | 115.7 | 84.3% | 521 | 13 |

Same RMSE as naive. Interval never missed, so Winkler 674 is just its width: way too wide.

# What I changed for the final file

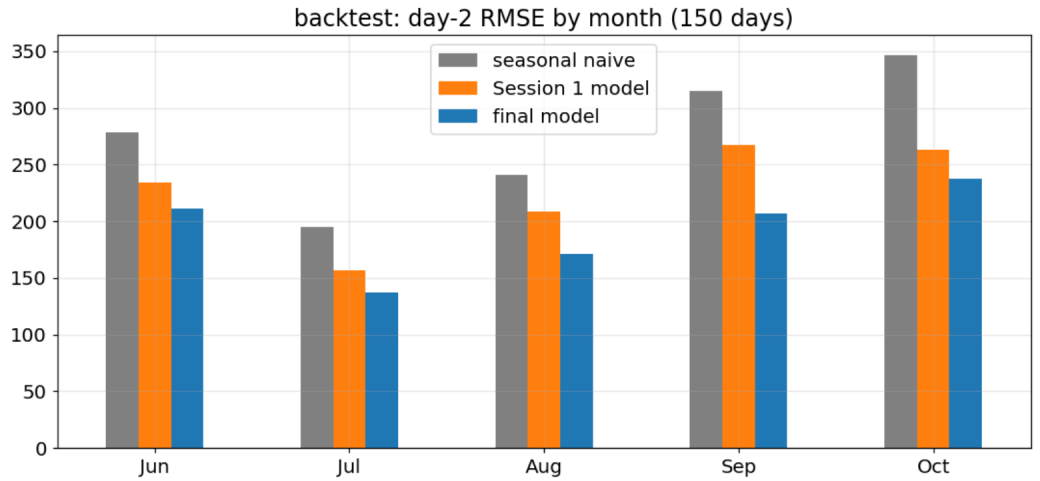
| problem in Session 1 | fix |
|---|---|
| backtest used real weather, the forecast file is noisy (more on day 2) | train and test on weather + noise (noise size estimated from the forecast file) |
| clock change shifts the sun by 1 hour | hour counted from the middle of daylight |
| growth: trees can't go above levels they saw | 2nd model predicts y / last 28-day mean, average of both |
| same interval width every day | width = a x (14-day volatility) + b x forecast |
| MAPE explodes on calm nights | small forecasts shrunk (x0.6 near zero) |
| leaderboard counts places, not averages | interval x0.9 (in backtest it wins about 2 days of 3) |

# Checking the final: 150-day backtest

**Day 2 only (hours 25-48), noisy weather**

| | naive | Session 1 | final |
|---|---|---|---|
| RMSE | 272 | 224 | 191 |
| MAPE | 225% | 145% | 89% |
| Winkler | 1850 | 1101 | 944 |
| coverage | 0.86 | 0.95 | 0.91 |

beats naive on 117 of 150 days




October is the hardest month for every model.

# Private leaderboard (28 Oct, hours 25-48)

| # | name | RMSE | MAPE | Winkler | score |
|---|---|---|---|---|---|
| 1 | Kristina Polonska | 77.8 | 48.2% | 490 | 2.0 |
| 2 | Mykola Utkin | 69.9 | 51.8% | 461 | 2.2 |
| 3 | Illia Narbut | 119.5 | 44.0% | 561 | 4.3 |
| 4 | Zakhar Huzii | 144.6 | 51.0% | 697 | 7.3 |
| 5 | Oksana Yefimova | 115.4 | 53.4% | 765 | 9.5 |
| 6 | Liliia Chervonetska | 133.3 | 71.6% | 722 | 11.0 |

- 3rd, with the final file
- best MAPE: the night shrink worked
- no misses, so Winkler 561 = width. Could be narrower
- weak point: RMSE, the daytime peak

Next time: better daytime peak, and a narrower interval when the last days are calm. Thanks!